

2020 AMC 10B Problem 2 - Solution

Review the full statement and step-by-step solution for 2020 AMC 10B Problem 2. Great practice for AMC 10, AMC 12, AIME, and other math contests

2020 AMC 10B Problem 2

Problem:

Carl has 5 cubes each having side length 1, and Kate has 5 cubes each having side length 2. What is the total volume of these 10 cubes?

Answer Choices:

- A. 24
- B. 25
- C. 28
- D. 40
- E. 45

Solution:

A cube with side length 1 has volume $1^3 = 1$, so 5 of these will have a total volume of $5 \cdot 1 = 5$.

A cube with side length 2 has volume $2^3 = 8$, so 5 of these will have a total volume of $5 \cdot 8 = 40$.

$$5 + 40 = \boxed{(E)45}$$

OR

The total volume of Carl's cubes is 5. This is because to find the volume of a cube or a rectangular prism, you have to multiply the height by the length by the width. So in this question, it would be $1 \times 1 \times 1$. This is equal to 1. Since Carl has 5 cubes, you will have to multiply 1 by 5, to account for all the 5

cubes.

Next, to find the total volume of Kate's cubes you have to do the same thing. Except, this time, the height, the width, and the length, are all 2, so it will be $2 \times 2 \times 2 = 8$.

Now you have to multiply by 5 to account for all the 5 blocks. This is 40. So the total volume of Kate's cubes is 40.

Lastly, to find the total of Carl's and Kate's cubes, you must add the total volume of these people. This is going to be

$$5 + 40 = \boxed{(E)45}$$

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